

Defeating Radiation Damping and Magnetic Field Inhomogeneity with Spatially Encoded Noise

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A simple NMR experiment capable of providing well resolved spectra under conditions where either radiation damping or static magnetic field inhomogeneity would broaden otherwise high-resolution NMR spectra is introduced. The approach involves using a strong pulsed magnetic field gradient and a selective radio-frequency pulse to encode a predetermined noise pattern into the spatial distribution of magnetization. Following readout in a much smaller field gradient, the noise sequence may be deconvolved from the acquired data and a high-resolution spectrum is obtained, eliminating the effects of

either radiation damping or the static field inhomogeneity. In the presence of field inhomogeneity a field map is also obtained from the same single transient. A quasi-two-dimensional version of the experiment eliminates the need for deconvolution and produces improved results with simplified processing, but without requiring a full two-dimensional experiment. Example spectra are shown for both radiation damping and one-dimensional field inhomogeneity with improvement in line-widths of more than a factor of 40.

1. Introduction

In many NMR experiments, achieving the highest possible resolution is important. In cases where proton densities are large, such as with protonated solvents, a phenomenon known as radiation damping can cause broadening and severe distortions of the solvent and nearby peaks. Radiation damping refers to the interaction between the nuclear spins and the probe detection circuit. As an in-plane component of nuclear spin magnetization precesses, it induces a voltage in the detection coil. The voltage gives rise to a current, which in turn creates a resonant oscillating field within the coil. This radiation damping field is phased so that it nutates the magnetization towards the static field (*z*) axis. When the angle between the static field and the magnetization vector is small enough ($\lesssim 60^\circ$), the decay of the transverse magnetization is approximated well by an exponential, but for larger displacements, a *sech*(*t*) time dependence is found.^[1] When the exponential approximation is valid, a broadened Lorentzian line is observed. For larger flip angles however, the lineshapes become severely distorted. Because the damping field is relatively small, it typically only directly affects a narrow bandwidth around the strong spectral line(s) of the concentrated solvent. Radiation damping may also be understood from the perspective of energy conservation. The current in the probe circuit produced by the precession of the nuclear spins dissipates energy. This energy must be obtained from the nuclear spins and is given up by the return of the spin magnetization to the *z*-axis.^[2]

Several methods are known to overcome radiation damping. One class of these techniques introduces severe resolution or sensitivity sacrifices. These methods include diluting the sample, detuning the probe, or deshimming the magnet. More sophisticated techniques include the application of correction pulses in between the acquired data points,^[3] the rapid dephasing and rephasing of the magnetization between data

points with magnetic field gradients,^[4,5] and the modification of the probe and receiver circuit design to actively cancel the induced current.^[6–8] Suppressing solvent signals^[9] provides another avenue of dealing with samples where radiation damping would cause problems. Most of these techniques required either delicate tuning of parameters to match the solvent properties or involved hardware modifications.

In contrast to radiation damping, magnetic field inhomogeneity broadens all the lines in an NMR spectrum. Again, several different techniques have been developed to allow high-resolution spectra to be obtained under these conditions. These techniques include the use of inter-^[10] or intra-^[11] molecular zero-quantum coherences, the Nuclear Overhauser Effect (NOE),^[12] or total spin coherences^[13] to produce a high-resolution one-dimensional NMR spectrum from a two-dimensional experiment. Single-transient pulse sequences have been developed, based on repetitive shaped radio-frequency (rf) and gradient pulses that directly compensate for the spatially dependent phase evolution in an inhomogeneous field.^[14,15] These techniques require mapping of the magnetic field distribution in advance and tailoring the correction pulses to the field distribution. Reference deconvolution is another method of producing sharp spectra from modestly inhomogeneous fields. In reference deconvolution, the lineshape observed for a well resolved spectral line is deconvolved from the spectrum to eliminate the effects of the field inhomogeneity. This technique can work well as long as the broadening to be removed is not too

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severe.^[16] The deconvolution of broad lines into sharp lines does tend to impose large signal-to-noise ratio penalties, as it involves the amplification of the signals and noise appearing late in the free-induction decay (FID). More recently Halse and Callaghan^[17] demonstrated a spatially resolved deconvolution technique that combines reference deconvolution with spatially resolved spectroscopy to spread the field inhomogeneity out over many voxels, so that each voxel bears only a portion of the line broadening. The linewidth improvement required by deconvolution for each voxel is then only due to the field inhomogeneity over the voxel, and the signal-to-noise penalty is reduced compared to traditional deconvolution of the entire inhomogeneity.

Herein we present a simple method of producing high-resolution spectra under conditions where radiation damping or a static field inhomogeneity would broaden the NMR spectra. The method is based upon the encoding of a deterministically structured pattern of noise into the spatial distribution of magnetization and was inspired by an optical spectroscopy technique known as noise autocorrelation spectroscopy with coherent Raman scattering (NASCARS),^[18] where a similar excitation-imposed noise structure is imprinted upon broad optical spectra to improve resolution.

2. Methodology: Spatially Encoded Noise

The experiment to be described is logically divided into two steps. First is an excitation step that prepares the sample's magnetization so that, within different transverse slices of the sample, the nuclear spin magnetization points along either the $\hat{x}$ or $-\hat{x}$ direction (Figure 1 b). The directions for the magnetization in the various slices are chosen according to a pseudo-

random noise sequence for reasons that are explained below. The phrase "spatially encoded noise" refers to this encoding of a noise sequence into the position dependence of the nuclear spin magnetization. In the second step of the experiment, a free-induction decay is acquired from this prepared sample in the presence of a weak magnetic field gradient. The weak gradient separates the signals from adjacent slices by an amount slightly larger than the linewidths expected in the spectrum. While this procedure would appear to greatly complicate the acquired spectrum, high resolution can in fact be recovered.

With a homogeneous sample in a homogeneous field, the spectra arising from each of the slices are identical, but shifted in frequency by the acquisition gradient and modulated in sign according to the encoding noise sequence. The acquired spectrum consists of the superposition of all of these individually well resolved, sign-modulated spectra. In the case of field inhomogeneity along the encoding direction, it is clear that the spectra from each of the slices individually will be better resolved than the ordinary spectrum from the entire sample volume, because the field variation over any one slice will be a small fraction of the total field variation over the whole sample. In the case of radiation damping, the resolution enhancement arises from the fact that at no time during the acquisition is there ever a large macroscopic magnetization vector. The magnetizations of the various slices, even when they are all "in phase" (along $\pm\hat{x}$) at the start of the acquisition, interfere with each other to reduce the voltage and corresponding current induced in the detection coil, resulting in a reduced radiation damping field. Because the radiation damping field is only able to damp lines nearby in frequency, the spread of frequencies due to the gradient limits the ability of the magnetization in one slice to contribute to the damping of magnetization in other slices.

The pulse sequence used is shown in Figure 1 a. To prepare the magnetization, a strong magnetic field gradient is applied to the sample along with a carefully crafted selective rf pulse. This pulse tips magnetization in different slices of the sample in different directions so that at the instant of the simultaneous spin echo and gradient echo (indicated by the vertical arrow), the magnetization lies, as desired, along either the $\hat{x}$ or $-\hat{x}$ direction in the rotating frame, as shown in Figure 1 b. Following the echo, the gradient field strength is reduced and a free-induction decay is collected. An idealized spectrum from a sample, the ordinary NMR spectrum of which consists of a single peak, is shown in Figure 1 c.

As noted above, this pulse sequence reduces the effects of radiation damping for two reasons. First, because the gradient is applied during the signal acquisition, the field produced by the magnetization from one slice of the sample has a reduced ability to affect the magnetization in other slices, because their resonant frequencies differ. Second, because the encoding of the magnetization is performed according to a pseudo-random noise sequence, the magnetizations in the various slices never add constructively, reducing the amplitude of the current and the corresponding radiation damping field induced in the detection coil. As long as the encoding gradient is large enough that the frequency range due to that gradient

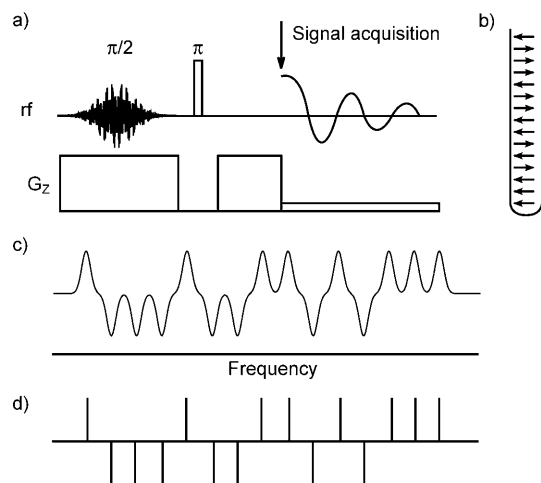


Figure 1. Spatially encoded noise. a) Pulse sequence used to encode the noise sequence into the magnetization's position dependence and read out the resulting spectrum. The selective pulse is the superposition of a series of amplitude- and phase-modulated, frequency-shifted Gaussians. The vertical arrow indicates the time of a simultaneous spin and gradient echo and marks the end of the encoding step and the beginning of the acquisition step of the experiment. b) Cartoon showing the arrangement of nuclear spin magnetization produced in the sample. c) Idealized spectrum. d) Stick spectrum that can be used to deconvolve the imposed noise lineshape to recover a high-resolution spectrum.

is much greater than the chemical shift range, a structure such as that shown in Figure 1c will be present for each peak in the NMR spectrum. However, the positions of these structures will be shifted according to their frequencies in the standard one-dimensional spectrum.

There are multiple ways of recovering a high-resolution spectrum from the data acquired in this fashion. In the NAS-CARS optical spectroscopy technique, high-resolution spectra are recovered from the autocorrelation function of the acquired spectra.^[18] The autocorrelation function, however, does not provide a standard spectrum, as peaks are present at frequencies corresponding to the differences between each pair of spectral lines. In our case, the most straightforward method of extracting a standard high-resolution spectrum is closely related to reference deconvolution.^[16] However, the deconvolution used herein is unique in that the reference to be deconvolved consists not of an inhomogeneity broadened lineshape, but rather of a series of infinitely sharp “sticks”, as shown in Figure 1d. Because the time-domain representation of the stick spectrum does not decay, the difficulty associated with amplifying noise late in the FID is, in principle at least, avoided.

In an inhomogeneous field, with the inhomogeneity for now confined to lie along the applied gradient direction, the positions of the peaks in the encoded noise structure will be shifted according to the local field in addition to the read gradient. A similar deconvolution may be used, but the relative positions of the peaks must be determined from the acquired spectra. If the inhomogeneity is not a simple linear dependence of field on position, the widths of the peaks as well as their positions may vary. In this case deconvolution by a simple stick spectrum will not produce an optimal result, as the stick spectrum assumes that the building blocks of the encoding are identically shaped peaks.

A second method by which a high-resolution spectrum may be recovered, without the necessity of deconvolution, makes use of the mathematical properties of maximal-length binary sequences (MLBSs).^[19] These sequences are well known in NMR, having been introduced some 40 years ago for “noise spectroscopy”.^[20,21] A property of these sequences which makes them useful both for noise spectroscopy and in the current experiment is that, if the sequence elements are assigned values $+1$ or -1 , the autocorrelation function of the sequence is given by Equation (1):

$$G_i = (N + 1)\delta_i - 1 \quad (1)$$

in which N is the number of bits in the sequence and $\delta_i = 1$ if $i=0$ and 0 otherwise. The autocorrelation is calculated for a cyclical sequence and the index i is the number of elements by which one copy of the sequence is shifted with respect to the other. For large N , the constant -1 term may be ignored, but for small values of N it is important. Given this autocorrelation function, it is evident that if a series of spectra are collected, each starting one step later in the cyclic MLBS, the set of resulting noise-encoded spectra can be separated into spatially resolved spectra simply by cross-correlating the series of raw

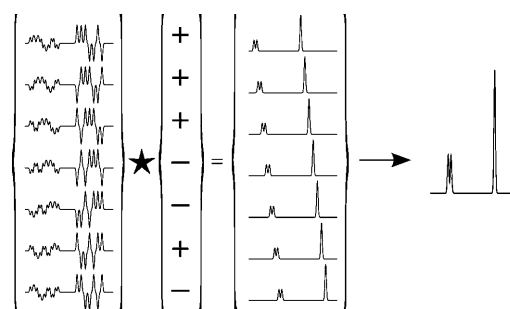


Figure 2. Reconstruction by cross-correlation of shifted MLBS excited spectra. The column on the left shows an idealized series of seven spectra generated using excitation pulses with sequential starting points of the MLBS: $\{+, +, +, -, -, +, -\}$. These spectra are then point-wise cross-correlated (indicated by the $\star$ operator) with the MLBS to yield the series of spatially resolved spectra in the middle. These spectra are then frequency shifted and co-added to yield the final spectrum on the right.

noise-encoded spectra with the MLBS. The resulting spatially resolved spectra can then be frequency-shifted and co-added to produce a final spectrum. This process is illustrated in Figure 2.

For this cross-correlation to work well, the constant offset -1 in Equation (1) must be considered. With sequence elements of ± 1 , the cross-correlation will leave parts of each peak scattered amongst the spatially resolved spectra. This may be remedied by multiplying the amplitude of the negative sequence elements by $a = 1/(1 \pm 2/\sqrt{N+1})$, as suggested by Yang et al.^[22] The $\pm$ indicates there are two choices for a , one greater than and one less than one. With such modified sequences, the autocorrelation function becomes [Eq. (2)]:

$$G'_i = \left(\frac{N+1}{2} + \frac{N-1}{2} a^2 \right) \delta_i \quad (2)$$

While the amplitude modulation required for these modified sequences could be incorporated directly into the selective pulse, the accuracy of the modulation would be sensitive to the exact pulse flip angle and would have the undesirable side-effect of leaving additional magnetization pointing along the z -axis. To avoid these issues, rather than modulating the amplitude of the negative sequence elements, we phase-modulate by a phase angle $\theta = \arccos a_{\pm}$ where we have chosen the positive sign of the $\pm$ in the definition of a to guarantee that $a < 1$. The experiment is then performed twice, once with the negative sequence elements phase-shifted by θ , and once by $-\theta$. The results of these two experiments are then added to yield the desired amplitude modulation, as illustrated in Figure 3.

In the case of radiation damping, the relative frequency shifts required of the spatially resolved spectra are known beforehand, because successive spectra are shifted by the spacing between peaks in the selective pulse (scaled by the ratio of the read-gradient strength to encode-gradient strength). In the presence of an inhomogeneous field, however, the spatially resolved spectra will also be shifted according to the local field. If the signal to noise in the individual spatially resolved

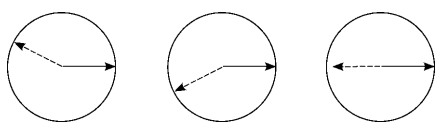


Figure 3. Illustration of the method used to amplitude modulate negative sequence components. The vectors indicate the location of the magnetization in the x - y plane of the rotating frame for the + (—) and - (----) sequence components. The experiment is performed twice with positive and negative phase shifts applied to the negative components. Co-adding results in the desired amplitude modulation in a pulse-flip angle insensitive fashion.

spectra is not sufficient to easily determine the necessary shifts, they may be found by cross-correlating each of the spatially resolved subspectra with one of the subspectra chosen as a reference. The peaks in these cross-correlation functions indicate the correct frequency shifts, and the shifts themselves provide a one-dimensional field map.

3. Results

The spectrum acquired with the pulse sequence shown in Figure 1a applied to a sample consisting of 87% H_2O , 5% D_2O , and 8% acetone is shown in Figure 4a. The 63-element-long encoding sequence is clearly seen for both the H_2O and acetone peaks in the raw spectrum. The noise structure corresponding to the water peak, expanded in Figure 4b, was fitted to a calculated spectrum consisting of 63 lines using the n2f nonlinear least-squares fitting routine from the AT&T port library.^[23] The positions, amplitudes, and phases of the lines were allowed to vary, and overall Gaussian and Lorentzian broadening factors were included to optimize the fit. The stick spectrum in Figure 4c shows the positions and amplitudes (but hides the $<30^\circ$ phase variations) of the peaks found in

the best fit. Much of the variation in the amplitudes of the sequence elements appears to be due simply to the variation in sensitivity with position in the coil that is observed in a one-dimensional image (Figure 4d). The stick spectrum was deconvolved from the raw spectrum in Figure 4a to produce the high-resolution spectrum shown in Figure 4f. This spectrum may be compared to the spectrum observed following a single small flip-angle ($\sim 22^\circ$) pulse in Figure 4e. The radiation-damped water line is well described as a Lorentzian having a full-width at half-maximum (FWHM) of 40 Hz, while the water line following deconvolution (Figures 4f,g) is more nearly a Gaussian with a FWHM of 1.16 Hz, a more than 34-fold decrease in linewidth. The Gaussian character of the resulting spectrum is expected from the Gaussian nature of the selective excitation pulses.

The noise in the baseline of the deconvolved spectrum arises from imperfections in the deconvolution and limits the dynamic range of the experiment. In Figure 4e, the apparent signal-to-noise ratio of the water peak is approximately 200. Replacing the 63-element MLBS with a 31-element-long sequence and cyclically rotating through all 31 possible starting points allows for reconstruction by cross-correlation rather than deconvolution. The results of this experiment are shown in Figure 5. The water linewidth here is 1.03 Hz and the reconstructed spectrum is much cleaner. While the deconvolution noise in Figure 4f is spread across the entire spectrum, imperfections in the cross-correlation reconstruction are confined to the regions near the strong peaks. In this case the ^{13}C satellites of the acetone peak, which are down in amplitude by more than a factor of 6000 from the water peak, are visible with a signal-to-noise ratio of about 10, indicating a dynamic range of the order of 60 000. Such a large dynamic range is not observed directly in the vicinity of the water peak, as residual

peaks appear at the positions of all the neighboring Gaussian peaks in the noise sequence. The most prominent of these are marked with vertical arrows in Figure 5c. The systematic pattern of amplitudes of these spurious peaks, positive on the up-field side and negative on the low-field side of the main peak, suggests that they arise due to a coherent effect rather than from imperfections in the implementation. These peaks indicate that the amplitude and/or phase of the solvent magnetization observed in one slice have components that are correlated with the encoded directions of the magnetization in the neighboring slices. No such systematic pattern of peaks is observed near the much weaker acetone peak (Figure 5d). We believe that these peaks arise from the residual effects of radiation damping, although distant dipolar field effects may also be important. Classical simulation of magnetization in one slice under the influence of radiation damping fields arising from both the observed slice and from a neighboring slice reveals that the phase of the magnetization does have a weak dependence on both the sign and relative frequency of the magnetization in the neighboring slice, as would be required to pro-

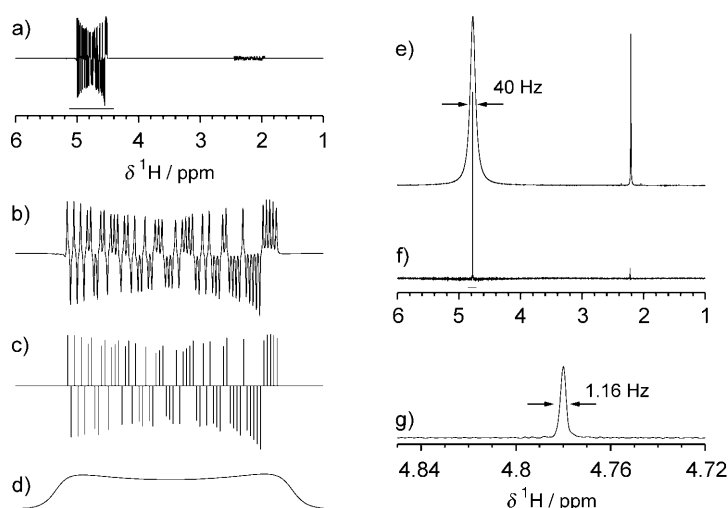


Figure 4. Spatially encoded noise to defeat radiation damping. a) Raw spectrum acquired from the pulse sequence of Figure 1a on a sample of water, acetone, and D_2O . The region shown underlined in (a) is expanded in (b) and may be compared to the stick spectrum used for deconvolution in (c). A one-dimensional image of the sample showing the variation in coil sensitivity with position is shown in (d). e) Spectrum acquired from a single small flip-angle pulse for comparison to the result of the deconvolution, shown in (f) with the underlined region shown expanded in (g).

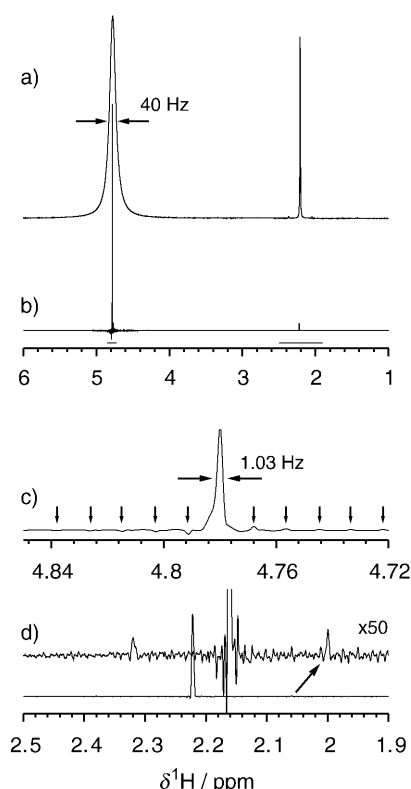


Figure 5. Quasi-2D experiment under radiation damping conditions. a) Result of a small flip-angle pulse on the water/acetone/D₂O sample showing the radiation damping broadened spectrum, for comparison to the resulting spectrum from the quasi-2D experiment where successive spectra are acquired with the MLBS encoding sequence cycled through all possible starting points and then reconstructed by cross-correlation. The full resulting spectrum is shown in (b), with the underlined regions shown expanded in (c) and (d).

duce the pattern of peaks observed. This suggests that these peaks may be reduced or eliminated by increasing the number of peaks in the noise sequence, at the cost of an increased number of steps required to cycle through the MLBS. Compared to the single-transient data of Figure 5, the attenuation of radiation damping in the quasi-2D experiment performed here is expected to be poorer because the number of sequence elements is smaller, resulting in reduced interference between the slices, and also because the readout gradient is 25% lower in this case.

Performance of the sequence in an inhomogeneous field is demonstrated with a sample of D-glucose in D₂O (Figure 6). The raw spectrum is shown in Figure 6a and an expanded view of the residual water peak is shown in Figure 6b. The stick spectrum found from a best fit to a sequence of 31 peaks having adjustable positions and amplitudes (but not phases, as allowing the phases to vary did not produce any appreciable improvement in the final spectrum) is shown in Figure 6c. The positions of the sticks were employed to create a field plot (Figure 6d). Deconvolving the stick spectrum from the raw spectrum provides the high-resolution spectrum

(Figures 6 f,g), which may be compared with the spectrum observed following a single hard pulse (Figure 6e).

Field inhomogeneity, due to deliberately mis-set Z, Z2, and Z3 shims, broadens the lines in the single-pulse spectrum over a range of about 42 Hz. Following the deconvolution however, the sharpest lines are well described as Gaussians with a FWHM of 0.86 Hz. The anomeric proton peaks at 5.24 ppm, which (in sucrose) are often used as a measure of field homogeneity, are approximately 1.4 Hz in width. Deconvolution noise again limits the dynamic range of the final spectrum. Both the field map and the reconstructed spectrum in Figure 6 come from the same *single-transient* experiment.

In this case rotating through the 31-element-long MLBS and reconstructing by cross-correlation produces the spectrum shown in Figure 7. Here, the anomeric proton FWHM is 1.35 Hz and the sharpest peaks in the spectrum are just over 0.9 Hz. The reconstruction required is essentially fully automatic and the frequency shifts used to align the spatially resolved sub-spectra immediately yield the one-dimensional field map (Figure 7d). The reconstructed spectrum of Figure 7 required a total of 62 transients to be collected.

4. Discussion

The results shown above demonstrate clearly that the noise-sequence-encoding strategy is able to produce high-resolution spectra in the presence of either radiation damping or one-dimensional static field inhomogeneity. In the case of radiation damping, the NMR signals from the strong solvent peaks persist nearly 40 times longer with the encoding than after a

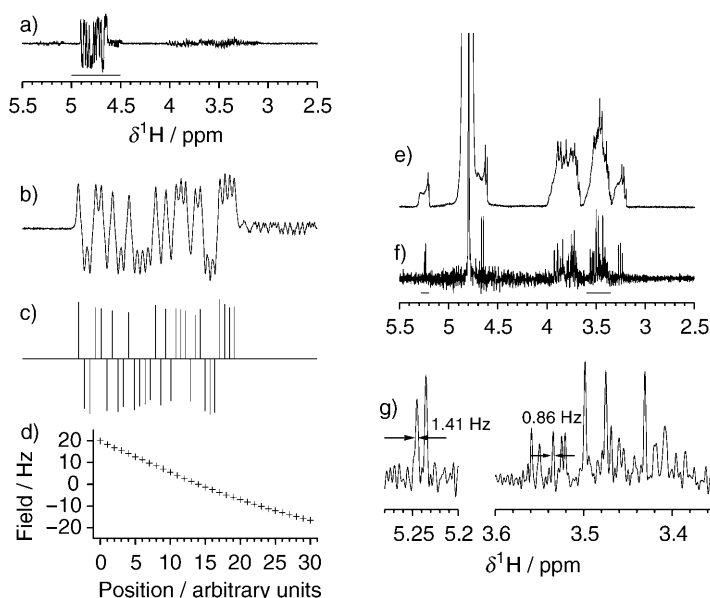


Figure 6. Spatially encoded noise in an inhomogeneous field. a) Raw spectrum acquired in a single transient using the sequence of Figure 1a on a sample of D-glucose in D₂O. The underlined region is shown expanded in (b), for comparison to the stick spectrum (c) used for deconvolution. d) Field map constructed from the stick positions in (c). A single pulse experiment yields the inhomogeneity-broadened spectrum in (e), to be compared to the result of the deconvolution of (c) from (a), shown in (f). The underlined regions are shown expanded in (g).

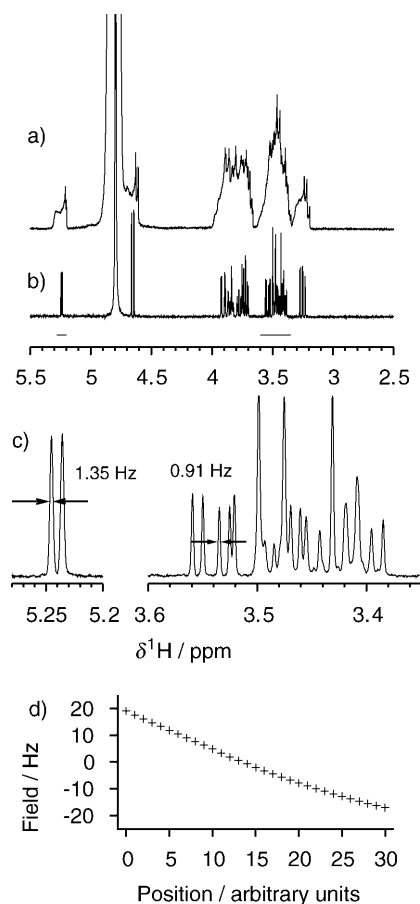


Figure 7. Quasi-2D experiment in an inhomogeneous field. a) Spectrum acquired from a D-glucose in D₂O sample acquired with a single pulse, for comparison to the high-resolution spectrum (b) recovered by cross-correlation from a sequence of 31 spatially resolved spectra acquired by cycling the MLBS encoding sequence through all possible starting points. The underlined regions are shown expanded in (c), and the field map constructed from the spatially resolved spectra is shown in (d).

simple hard pulse. Residual radiation damping (combined possibly with distant dipolar field effects) does appear to introduce some distortions in the water/acetone spectra. Both the phase-shifts required to adequately fit sticks to the one-dimensional encoded spectrum and the spurious peaks found following the quasi-2D reconstruction do not appear to affect the glucose/D₂O sample where radiation damping and distant dipolar fields are significantly reduced.

The encoding gradients used spread the resonances out over a frequency range of ~250 kHz, which may be a nontrivial fraction of the probe resonance bandwidth, especially if a high-Q cryoprobe were employed. In principle, this wide frequency range could contribute to amplitude and phase variations amongst the excited slices and these variations would depend on probe tuning. It seems unlikely that this issue contributes appreciably to the distortions discussed above, as they were only observed for the concentrated solvent signals. Any such variations, however, would be handled in the single-transient reconstruction by allowing the phases and amplitudes of the sticks to vary, as was done for the spectrum in Figure 4. In

the quasi-2D reconstruction strategy, the net result of probe bandwidth limitations and mis-tuning would be that the spatially resolved spectra would show phase and amplitude variations with respect to each other. However, each subspectrum would be a faithful representation. Co-adding these subspectra without phase adjustment would result in a slight decrease in the signal-to-noise ratio, but would not cause any other distortions.

The spectra resulting from the reconstructions display a modest T_2 weighting in the amplitudes of the various peaks due to the use of a spin echo, but are otherwise quantitative, as long as the encoding gradient strength produces a frequency range much larger than the chemical shift range. The linewidths of the resulting spectra may include contributions from both the Gaussian nature of the excitation pulses, as well as from translational diffusion in the acquisition field gradient. In all of the spectra shown herein, the latter contribution is negligible, while the former contributes a sizable fraction of the linewidths. The observed linewidths result from convolution of the natural (T_2) linewidths with the Gaussian excitation profile. Differences in linewidths amongst the peaks in a spectrum are physically meaningful and generally indicate differences in T_2 .

In comparison to previous techniques developed to eliminate the effects of radiation damping, this technique is simple to implement and is robust with respect to the solvent spectrum (single or multiple peaks), solvent concentration, probe tuning, and rf power fluctuations. It requires a relatively high-performance pulsed-field gradient and does not appear to be able to produce as high a dynamic range close to solvent peaks as some previous techniques. It is likely that the implementation can be optimized to increase the dynamic range over what was demonstrated herein. In comparison with the NASCARS optical spectroscopy technique,^[18] the noise sequences used here are very short. This is in part because longer sequences are required to achieve a useful dynamic range with the autocorrelation reconstruction strategy used there. It is likely, however, that the suppression of radiation damping, especially in the quasi-2D version of this experiment, can be improved by increasing the sequence length.

In an inhomogeneous field, the pulse sequence shows the unique feature of being able to provide not only a high-resolution spectrum in a single transient, but also a field map. Again, with reconstruction by deconvolution from a single scan, the dynamic range is limited by imperfections in the deconvolution, but by cycling through the MLBS a clean and distortion-free reconstruction is obtained. In comparison to other methods of obtaining high-resolution spectra in inhomogeneous fields, most of which rely on a two-dimensional experiment, this technique has the advantage of being very fast, because even for the quasi-2D version (where cycling through an MLBS is used), the number of acquisitions required is set by the length of the MLBS rather than by the ratio of spectral width to resolution, as it is in an experiment where a high-resolution spectrum is acquired in an indirect dimension. In comparison to single-transient techniques where periodic correction pulses are used, this experiment is very simple to implement and requires no field mapping or tailoring of the sequence to be

done beforehand. As demonstrated however, it has the disadvantage of being able to eliminate field inhomogeneity in only one spatial dimension. Approaches to extend the method to inhomogeneity in more spatial dimensions are discussed below.

For larger inhomogeneities, the ultimate resolution in the final spectrum will be limited by the spreads in resonant frequencies over the individual excited slices. For larger inhomogeneities then, one would want to reduce the slice thickness, either by leaving larger gaps between them at the cost of sensitivity, or by increasing the total number of slices. Ultimately, the maximum inhomogeneity that can be handled is set by the requirement that the encoding gradient strength needs to be much larger than the gradients associated with the inhomogeneity, although practical limits on the number of slices that may be faithfully encoded might be more constraining. Line broadening due to diffusion in the inhomogeneity (or acquisition) gradient may also limit the final resolution in very inhomogeneous fields. We believe that inhomogeneities of up to a few hundred Hz can be handled practically with the technique.

4.1. Sensitivity

Acknowledging that the sensitivity of the technique, when spectral reconstruction is based upon deconvolution of a single transient, is severely limited by the accuracy of the deconvolution, we focus on the sensitivity expected when spectral reconstruction is undertaken with the quasi-2D strategy.

Because the preparation step of the pulse sequence only excites magnetization in discrete slices of the sample and these are spread out in frequency, the method described does involve a loss in sensitivity. In the glucose experiments shown above, which were optimized for resolution rather than sensitivity, the excitation pulses excited approximately 17% of the magnetization into the x - y plane. This magnetization is divided amongst 31 discrete peaks that must then be co-added. Compared to an ordinarily acquired one-dimensional spectrum in a homogeneous field using the same number of transients, we expect to find the signal-to-noise ratio reduced by a factor of approximately $\sqrt{31}/0.17 \approx 33$. In practice, the signal-to-noise ratio of the glucose spectrum in Figure 7, which required 62 transients, is a factor of ~ 3.7 lower than that of a single-transient spectrum, acquired in a homogeneous field so that the linewidths were comparable. Normalized to the same 62 transients, we find in this experiment a signal-to-noise ratio about 29 times lower than straightforward excitation (in a homogeneous field). This penalty is somewhat less severe than the factor of 33 expected, almost certainly due to the fact that the receiver gain could be increased for the noise-encoded spectra.

One portion of the signal-to-noise penalty, arising from the fact that only a portion of the signal's magnetization is excited by the selective pulse, can most certainly be reduced by further optimization of the experimental conditions. In fact, in the spectrum of Figure 4a, ca. 33% (rather than 17%) of the magnetization is excited and so the penalty there is reduced by a

factor of two. The second part of the penalty, which derives from the separation of the signal into different subspectra, is common to any resolution-enhancement technique that divides spectral intensity amongst slices in a second dimension. For example, the technique of Halse and Callaghan^[17] divided signals over as many as 128 voxels and so a sensitivity loss of $\sqrt{128} \approx 11$ is expected from this aspect of that strategy. This value does not include further losses due to their deconvolution step and does not account for the time necessary to acquire reference spectra. On the other hand, techniques involving correction pulses between data points generally require an increase in noise bandwidth, which brings its own sensitivity penalty.

In the case of radiation damping, the signal-to-noise ratio appears to be further limited in the regions close to the strong peaks by the distortions discussed earlier. Further experimental and theoretical studies to fully understand the limitations of the technique and their implications on the signal-to-noise ratios observed under radiation damping conditions will be undertaken and published elsewhere.

4.2. Inhomogeneity in Three Dimensions

There are many possible ways of extending the method to handle static field inhomogeneity in three dimensions. Perhaps the simplest is that shown in Figure 8a. No change is required to the encoding scheme and the readout is essentially a three-dimensional imaging sequence with phase encoding in two dimensions. If the field inhomogeneity across any transverse slice of the sample is not too great, it should be possible to use a relatively small number of phase-encoding steps in the in-plane directions, perhaps of the order of 8×8 , so that the readout in the phase-encoding directions would not require too many transients. Following Fourier transformation in the two indirect dimensions, a deconvolution procedure similar to that used in the pure one-dimensional case (Figure 6) would be undertaken for each pixel in the plane. The stick spectrum required would be slightly different for each pixel. This three-dimensional experiment would provide a high-resolution spectrum and three-dimensional field map.

The reconstruction required from the scheme shown in Figure 8b would be somewhat simpler, as the spatial information in all three dimensions is encoded in the magnetization with selective pulses rather than by phase encoding. Three separate noise-encoding steps are carried out, one each for gradients in each spatial dimension. In Figure 8b, the first two selective pulses are π pulses, where positive sequence elements invert the magnetization in a slice and negative elements leave the magnetization along $\hat{z}$. The final selective pulse is identical to that in Figure 1a. Spectra would then be collected with the first two encoding pulses cycled through their respective sequences and the spatial resolution in the plane achieved by cross-correlation with the appropriate binary sequence. In this case one need not be restricted to maximal length binary sequences, and in fact, other types of binary sequences for the in-plane encoding (e.g. Hadamard^[24]) would likely be preferred. This and other types of encoding sequence are discussed fur-

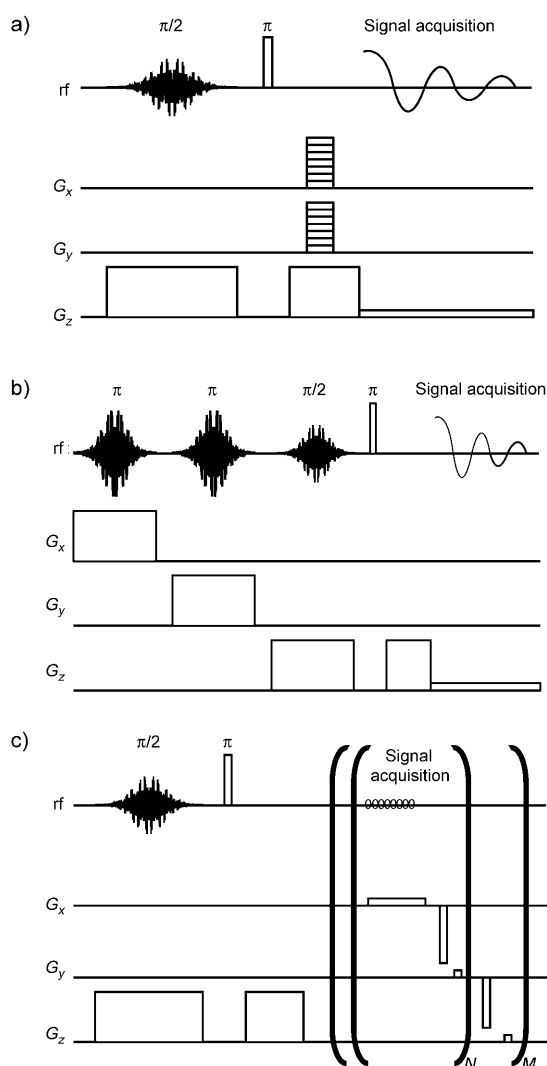


Figure 8. Sequences for collecting high-resolution spectra in fields with three-dimensional inhomogeneity. a) Phase-encoding gradient pulses are added to the sequence of Figure 1a for the $\hat{x}$ and $\hat{y}$ directions. b) Phase encoding replaced by binary sequence-encoding pulses. c) Acquisition of (a) is compressed into a single-transient experiment.

ther below. Alternatively, the encoding could be cycled through the relevant sequences in all three dimensions so that the reconstruction would require no fitting, just cross-correlations in each of the three dimensions. This however would mean a substantial increase in the number of transients required.

To collect high-resolution spectra in a single transient, the sequence shown in Figure 8c could be used. This is essentially the same sequence as that in Figure 8a, but with the acquisition compressed into a single transient in the style of echo-planar imaging (EPI). While this acquisition sequence might appear to be quite demanding on the gradients, it is likely that the number of in-plane points required will be relatively small. Many other options are possible, for example a hybrid method using regular phase encoding in one direction with EPI-like detection in the other two, or sequence rotation could be employed in one or two dimensions.

4.3. Time-Dependent Fields

One application where magnetic field homogeneity is often an issue is at the very highest field strengths in resistive or resistive/superconducting hybrid magnets. In these systems, not only is the field often inhomogeneous, but temporal field fluctuations at frequencies far exceeding feasible field-frequency-lock bandwidths may also be problematic.^[10,25] The spatially encoded noise strategy described herein can be adapted to handle fluctuating fields in addition to static field inhomogeneity. Field fluctuations will affect all of the spectral lines identically and so their effects would appear to be amenable to deconvolution, and in fact deconvolution of such fluctuations has been demonstrated.^[25,26] In order to carry out such a deconvolution knowledge of the field as a function of time is required during signal acquisition. Gan et al.^[25] obtained this information by collecting ^1H and ^2H spectra simultaneously while Iijima et al.^[26] used the induced electromotive force in a pick-up coil. One of these approaches could be employed here, where each transient would be deconvolved individually before any cross-correlation or other deconvolution steps.

4.4. Other Noise Sequences

In the demonstrations shown, the only noise sequences used are MLBSs but other types of sequences are worth considering. Some candidates are sequences based upon rows or columns of Hadamard matrices^[24] and Frank^[27,28] (or Chu^[29,30]) sequences which were recently introduced to NMR by Blümich et al.^[31] For the elimination of radiation damping, neither of these choices are likely to perform well unless applied very carefully. Hadamard sequences often contain repetitive structures and so are unlikely to be as effective as the MLBSs in attenuating radiation damping due to periodic partial refocusing of the magnetization in different excitation slices. Because the structures are different for different rows in a Hadamard matrix, the severity of this problem would vary for the different rows, making these sequences unsuitable for the quasi-2D version of the experiment described. Frank and Chu sequences contain a well-defined structure that is essentially that of a discretized frequency sweep. Thus, if applied in the most straightforward fashion, with the sequence elements encoded in space in sequential order, full periodic refocusing can be expected. If the order of the sequence elements were scrambled before encoding however, these sequences are likely to work well in both the single-transient and quasi-2D versions of the experiment.

For the first and second encodings in a sequence such as that in Figure 8b, the Hadamard-matrix-based sequences are likely preferred, as they possess binary orthogonality without requiring any sort of amplitude or phase modulation, thus they lend themselves well to the π pulses used.

4.5. Multi-Dimensional NMR

The basic noise-encoding pulse sequence of Figure 1a is suitable for use as a building block in many types of multi-dimensional NMR experiments. For example, a noise-encoded

COSY^[32] experiment could be performed by inserting, at the position of the simultaneous spin and gradient echo in Figure 1a, a t_1 period (with G_z off) and a hard $\pi/2$ pulse. Signals would then be acquired as above with the weak acquisition gradient. A 2D J-resolved spectrum^[33] could be collected by inserting a $t_1/2 - \pi - t_1/2$ block in between the encoding and acquisition periods instead. Incorporation of the noise encoding/decoding elements into many other kinds of multi-dimensional experiments is similarly straightforward. In processing the spectra acquired in this way, the deconvolution would need to be performed for each slice in the direct dimension. The limited dynamic range due to imperfect deconvolution would be at least partially alleviated in such experiments because deconvolution noise would be limited to one-dimensional slices containing strong peaks. In a time-dependent field, deconvolution of the field's time dependence could again be performed on a transient-by-transient basis before any other processing.

5. Conclusions

We have demonstrated a simple NMR pulse sequence that is able to produce high-resolution spectra in the presence of both radiation damping and inhomogeneous magnetic fields, based upon the encoding of a pseudo-random noise sequence in the spatial distribution of the nuclear magnetization. With this technique, high-resolution spectra may be acquired in a single transient and in cases of field inhomogeneity, the same single transient also yields a field map. Imperfections in the deconvolution required for the single-transient experiment limit the achievable dynamic range, but a large improvement may be obtained, and the need for deconvolution eliminated, by collecting a series of spectra with the noise sequence cyclically shifted.

The noise-encoding strategy introduced herein is amenable to incorporation into more complicated multi-dimensional NMR experiments and may be useful in the collection of high-resolution spectra from very high-field resistive or hybrid magnets where field fluctuations also contribute to broadening or lineshape distortion.

Experimental Section

All measurements were made on a Varian Unity Inova 400 spectrometer with an indirect detection probe. The probe has the capability of producing a single field gradient along the main field axis with a maximum gradient strength of ~ 54 Gauss cm^{-1} . The gradient amplifier is driven by a 16-bit digital-to-analog converter. The selective pulses were constructed as superpositions of frequency shifted, amplitude and phase modulated Gaussian peaks. The selective pulses each contained 4096 discrete time steps, which were delivered over pulse durations of 2.048 or 4.096 ms. The waveform generators on the spectrometer are capable of 0.5° phase resolution and 10-bit amplitude resolution. The total echo delay, measured from the center of the selective pulse to the center of the echo, was either 3.048 or 5.096 ms.

In all cases, the gradient during the selective pulse spread the resonance frequency out over ~ 250 kHz and was then reduced by a factor of either 1250 or 1667 during signal acquisition.

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